

Goal. Practice evaluating integrals using substitution and/or integration by parts.

1. Evaluate the following integrals. Write down which integration technique you use.

(a) Show that $\int_1^2 x^2 e^x \, dx = e(2e - 1)$.

Hint: Do integration by parts twice.

(b) Evaluate $\int x^3(x^4 - 17)^{24} \, dx$

Hint: You may use substitution or integration by parts. Which one is easier?

(c) Evaluate $\int \ln(x) \, dx$.

Hint: For any function $f(x)$, $f(x) = 1 * f(x)$.

(d) Use the substitution $u = 25 - x^2$ to evaluate $\int \frac{x^3}{\sqrt{25 - x^2}} \, dx$.

Hint: If $u = 25 - x^2$, what does x^2 equal?